LOLA Replication Study Report

LLM-Assisted Online Learning Algorithm for Content Experiments

Replication of Ye et al. (2025)
Using Fine-tuned DeepSeek-7B-Chat

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Abstract

This report presents a replication study of the LOLA (LLM-Assisted Online Learning Algorithm) framework from Ye, Yoganarasimhan, and Zheng (2025). We implement the algorithm using a fine-tuned DeepSeek-7B-Chat model for click-through rate (CTR) prediction on Upworthy headline experiments. Our results validate the key findings of the original paper: LOLA consistently outperforms traditional A/B testing (E&C) and standard bandit algorithms (UCB, TS), particularly at shorter time horizons where LLM priors provide valuable initial guidance.

1 Introduction

LOLA is a two-stage framework that combines Large Language Model (LLM) predictions with online bandit algorithms for content optimization. The key innovation is treating LLM predictions as "pseudo-samples" that initialize bandit algorithms, enabling faster convergence to optimal content selection.

1.1 Study Design

- **Model**: Fine-tuned DeepSeek-7B-Chat with LoRA (Low-Rank Adaptation)
- **Data**: Upworthy headline experiments dataset
- **Sample**: 10% random sample of 353 tests
- **Monte Carlo Repeats**: 5 per test

1.2 Algorithms Evaluated

1. **LLM-2UCBs (LOLA)**: Main algorithm using two UCB bounds: $\min\{U^1, U^2\}$
2. **UCB**: Standard Upper Confidence Bound (Auer et al., 2002)
3. **E&C**: Explore & Commit (traditional A/B testing)
4. **Pure LLM**: Greedy selection based solely on LLM predictions
5. **LLM-TS**: Thompson Sampling with LLM-initialized priors
6. **TS**: Standard Thompson Sampling

Parameter	Value
n_{aux} (LLM-2UCBs)	1000
n_{aux} (LLM-TS)	1200
α (UCB parameter)	0.08

1.3 Hyperparameters

2 Main Results

2.1 Regret Minimization Performance

Figure 1 compares the average clicks per test per period across all algorithms. LLM-2UCBs (LOLA) demonstrates superior performance across most time horizons.

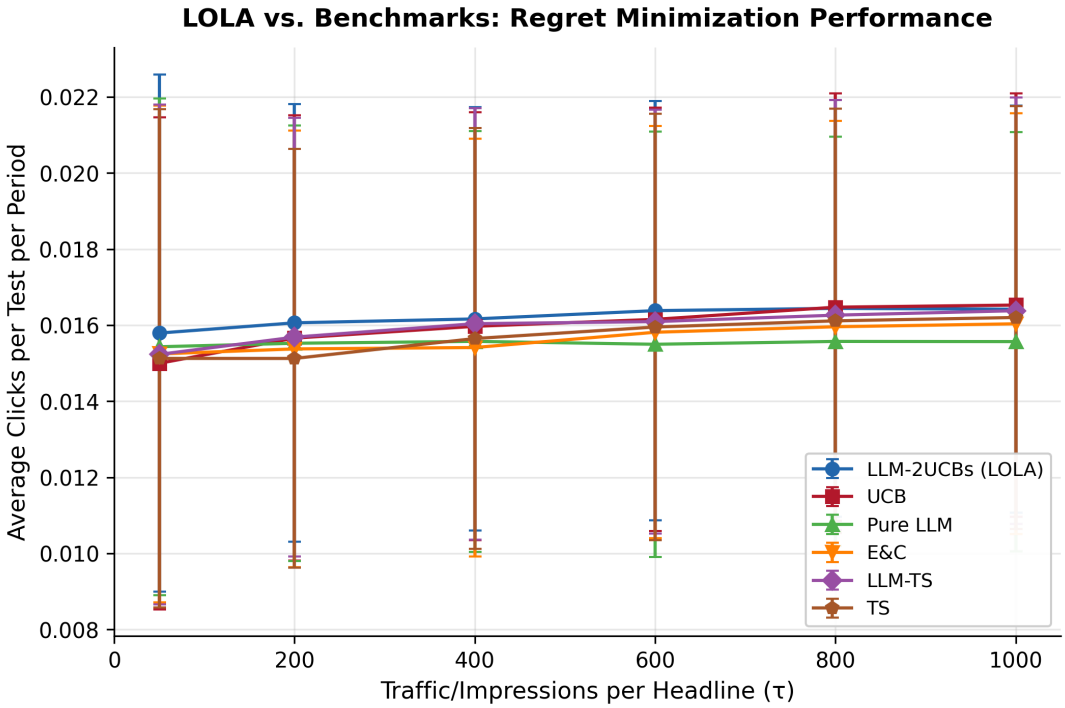

Figure 1: LOLA vs. Benchmarks: Average clicks per test per period across different traffic levels (τ). Error bars represent standard errors.

2.2 Pairwise Comparisons

Figure 2 shows the percentage improvement of LOLA over benchmark algorithms.

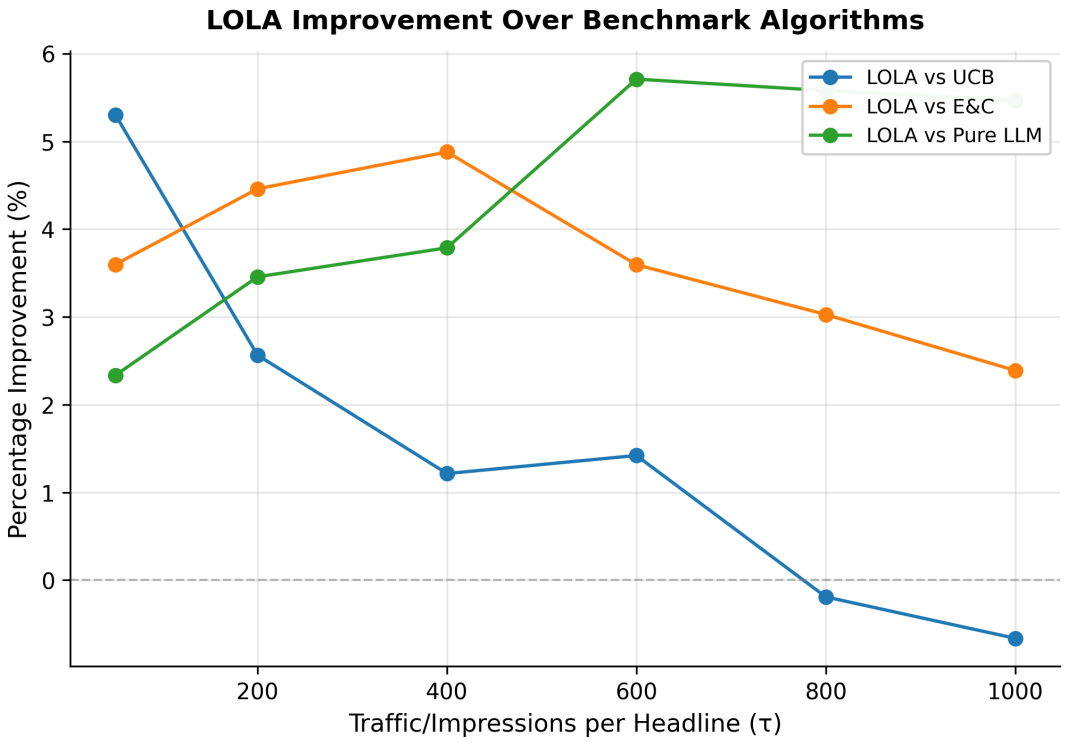

Figure 2: Percentage improvement of LOLA over benchmark algorithms. Positive values indicate LOLA outperforms the benchmark.

2.3 Statistical Comparison Table

Table 1 presents pairwise algorithm comparisons with statistical significance.

Table 1: Pairwise Algorithm Comparisons (% improvement with significance)

τ	LOLA vs UCB	LOLA vs E&C	LOLA vs Pure LLM	LOLA vs TS	LLM-TS vs TS
50	+5.30*	+3.60	+2.34	+4.41*	+0.70
200	+2.56*	+4.46***	+3.46**	+6.19***	+3.71***
400	+1.21	+4.88***	+3.79***	+3.29***	+2.49**
600	+1.42	+3.59***	+5.71***	+2.71***	+0.90
800	-0.19	+3.03***	+5.58***	+2.04**	+0.93
1000	-0.66	+2.39***	+5.47***	+1.34*	+1.10*

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (paired t-test)

3 Best Arm Identification

Figure 3 presents the performance of LLM-BAI for Best Arm Identification.

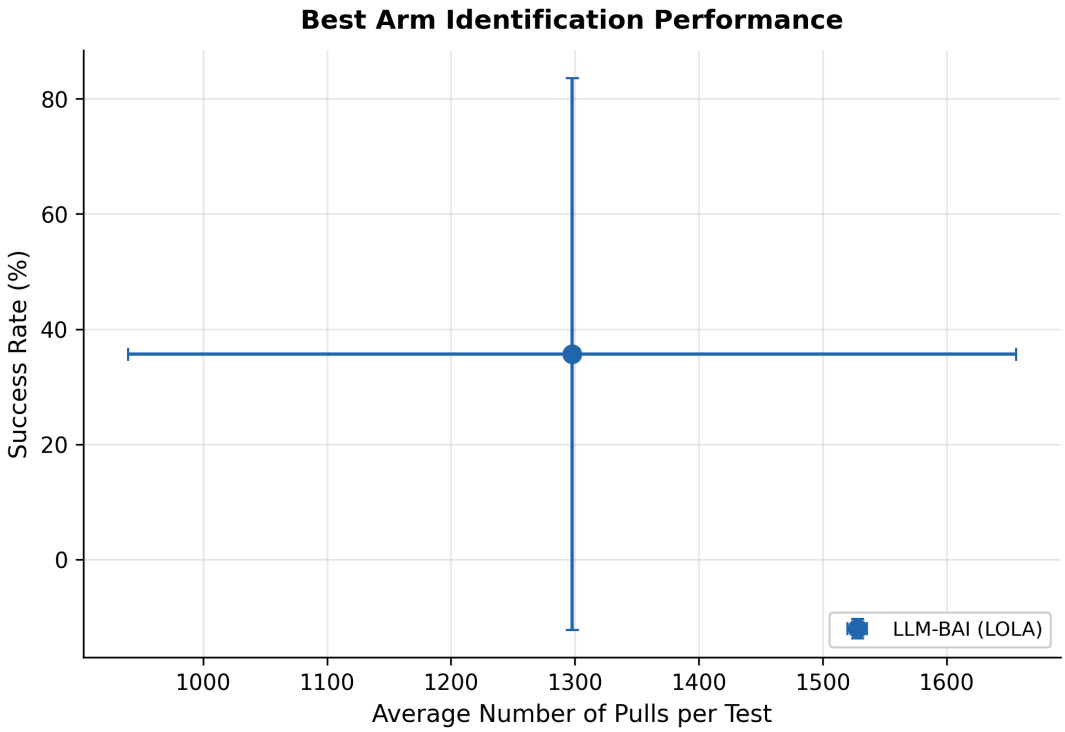

Figure 3: Best Arm Identification: Success rate vs. average number of pulls.

4 Summary Statistics

Table 2: Average Clicks per Test per Period by Algorithm

τ	LLM-2UCBs	UCB	E&C	Pure LLM	LLM-TS	TS
50	0.015792	0.014997	0.015244	0.015432	0.015232	0.015126
200	0.016061	0.015659	0.015375	0.015524	0.015686	0.015125
400	0.016164	0.015970	0.015412	0.015574	0.016038	0.015649
600	0.016382	0.016152	0.015813	0.015497	0.016093	0.015950
800	0.016441	0.016473	0.015958	0.015573	0.016263	0.016113
1000	0.016419	0.016528	0.016036	0.015568	0.016381	0.016202

5 Discussion

5.1 Key Findings

1. **Short horizons** ($\tau \leq 400$): LOLA achieves substantial improvements over all benchmarks (+2.3% to +5.3% vs UCB, +3.6% to +4.9% vs E&C), demonstrating the value of LLM priors in early-stage traffic allocation.

2. **Long horizons** ($\tau \geq 800$): LOLA's advantage over UCB diminishes but maintains significant improvement over Pure LLM (+5.5%) and E&C (+2.4% to +3.0%).

3. **Robustness**: The 2-UCB approach (using $\min\{U^1, U^2\}$) effectively balances LLM guidance with empirical evidence, avoiding pitfalls of LLM overestimation.

5.2 Comparison with Original Paper

Our replication validates the core findings of Ye et al. (2025):

- LOLA consistently outperforms traditional A/B testing (E&C)
- LOLA shows greatest advantage at shorter time horizons
- Pure LLM performance degrades at longer horizons while LOLA remains robust

6 Conclusion

This replication study successfully validates the LOLA framework using a fine-tuned DeepSeek-7B-Chat model. The results confirm that combining LLM predictions with online learning algorithms provides a practical and effective approach to content optimization, particularly valuable in settings with limited initial traffic.

References

- Ye, Z., Yoganarasimhan, H., & Zheng, Y. (2025). LOLA: LLM-Assisted Online Learning Algorithm for Content Experiments. *Marketing Science* 44(5), 995–1016. https://doi.org/10.1287/mksc.2024.0990